EART27201— Sedimentary Rocks and Fossils
Fossil Practicals — Brachiopods & Molluscs

Russell Garwood | russell.garwood@manchester.ac.uk

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https://www.russellgarwood.co.uk/EART27201.html

Under each QR code in this booklet is a keyword in italics - click the associated link on the page above, and it will take you to the correct target.

Welcome… To the wonderful world of brachiopods and molluscs. It's great to have you here. Today we're going to learn a bit more about the fossil groups that we met in today's lecture - let's start with brachiopods.

Brachiopods

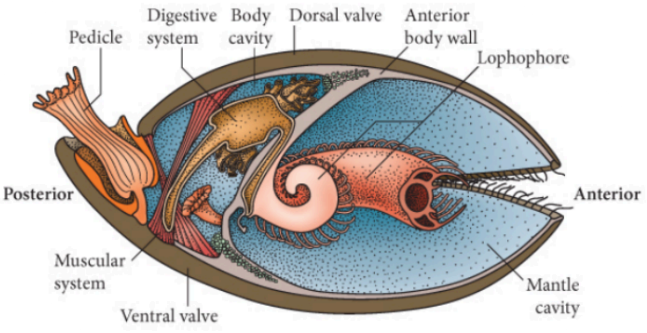
We learned in the lecture that there are three major groupings of brachiopods in the latest taxonomy of the group. Let's meet these a little more detail now (well, those we have available as fossils).

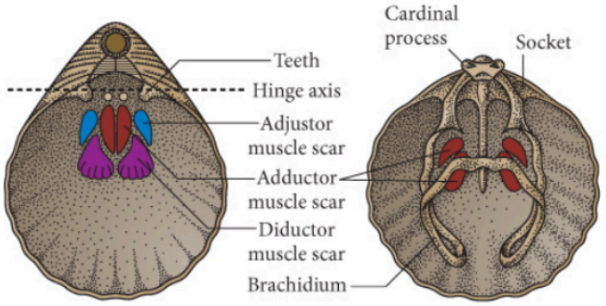
Rhynchonelliformea: This group represents the greatest diversity of brachiopods today and in the fossil record, and maps to the original idea of the Articulata. It comprises brachiopods with a pair of calcite valves and a posterior hinge (i.e. they pivot open and closed at the back of the animal). The calcite in their shell is distinct from the other groups of brachiopods in having a layer of fibrous calcite. The shells open anteriorly along the commissure, which zig-zags in some species increasing the area of the gape but not its width. This, and associated ridges, also strengthen the shell. Some members also have a depression in the middle (the median fold and sulcus) to separate incoming from outgoing water. The hinge (or articulation) is achieved using teeth on the ventral valve corresponding to sockets on the dorsal one. The shells open and close through muscle action - the diductor muscles open the shell, and adductors close it. These leave scars in the interior of the shell. Internally, some species have a structure called the brachidium supporting the lophophore, and the group lacks an anus (they just shoot waste out of their mouth). Most (but not all) members of this group attached to substrate using a pedicle that emerged through an opening (the foramen) in the ventral valve. The morphology of the group is summarised in the images below (courtesy of Benton & Harper 2020), and you can learn more on the link to the right.

You can explore the difference between the three types of brachiopods using 3D models on the website associated with brachiopods:

Link *Brachiopods*

Rhynchonelliformea

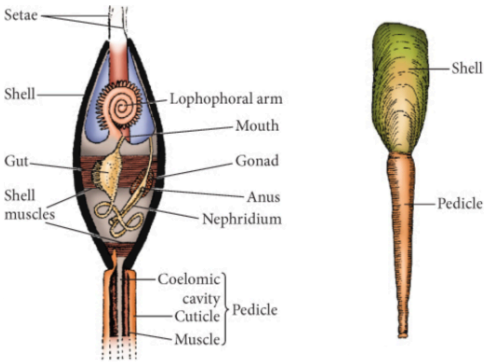




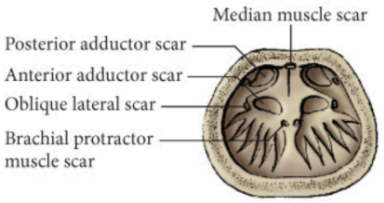
Linguliformea: These little guys don't have calcite shells. Rather their shells are organophosphatic (comprising calcium phosphate, proteins and a natural polymer called chitin) and are typically fairly symmetrical. They open and close these using a complex system of muscles, without an associated hinge mechanism. In brief, opening is achieved through the withdrawal of the soft parts posteriorly forcing the valves apart. Relaxing then allows the valves to close again. The pedicle in some members of the group emerges from between the valves, although others do have a foramen as in the above group. These animals have a recognisable U-shaped alimentary tract ending in an anterior anus. Modern forms are burrowers specialising in inter- to subtidal habitats.

Craniiformea: This is a diverse group. They are inarticulate (i.e. lack a hinge, like the last group), but have calcite shells like the first group. This is sometimes described as organocarbonate - the shells are constructed from an outer organic layer, and then two layers of calcite. These are punctate: i.e. they have small canals of tissue through the shell. The valves open and close through hydraulic mechanisms, rather than employing muscles. In general, the group attaches to a hard object with all or part of the ventral valve, and as such, individuals do not have a pedicle.

Linguliformea

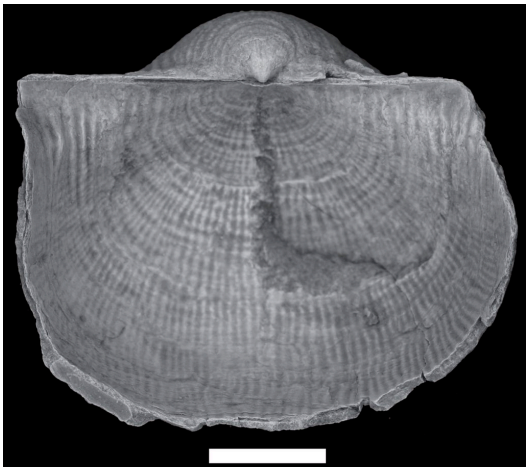


Craniiformea

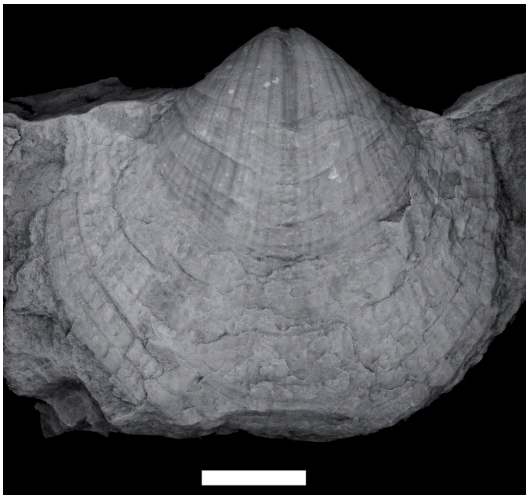


Brachiopod task 1: There are a couple of brachiopods in front of you, *Productus* and *Spirifer*. These are fairly typical Palaeozoic brachiopods, preserved closed with both valves, so you can't see much of the internal detail shown in the diagrams above. In general, palaeontology has a bit of a reputation for practicals being based on drawing specimens to help get to learn their morphology. This isn't my jam - but I do want you to think about the animals and their anatomy. To that end, I've pasted images of the two genera below. Using the hand specimens to guide you, label the important anatomical directions (dorsal and ventral valves, if you can see them; anterior and posterior) You can also see growth lines, and ridges on both, and a foramen on one. On the hand specimen make sure you are happy with what the commissure is. If you would actually rather draw and label the one in front of you, Russell can provide some paper!

Productus sp. Brachiopod; Upper Carboniferous (Pennsylvanian); Seminole Formation; Tulsa County, Oklahoma - photo by Sam Noble Museum (CCBY)



Sprifier sp.Brachiopod; Upper Carboniferous (Pennsylvanian); Bloyd Formation; Mayes County, Oklahoma - photo by Sam Noble Museum (CCBY)



Which of our three major divisions of the brachiopods do these represent, and importantly, *why*? Justify your answer.

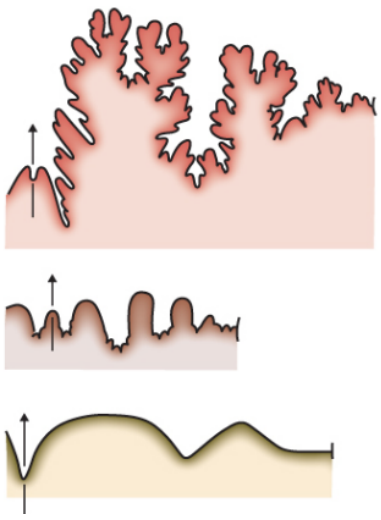
Let's have a little foray into paleoecology - i.e. use the anatomy of these animals to think about how they may have lived in life. I'm not giving much away when I say that brachiopods are/were filter feeders, which often attach to the substrate, or lie on the sediment. But we can go a bit further than this - the image on the next page summarises different modes of life within the brachiopods.

Brachiopod task 2: Using this image, and thinking about the morphology of the two species you have labelled for the last exercise, what do you think the mode of life was for each species? Justify your answer.

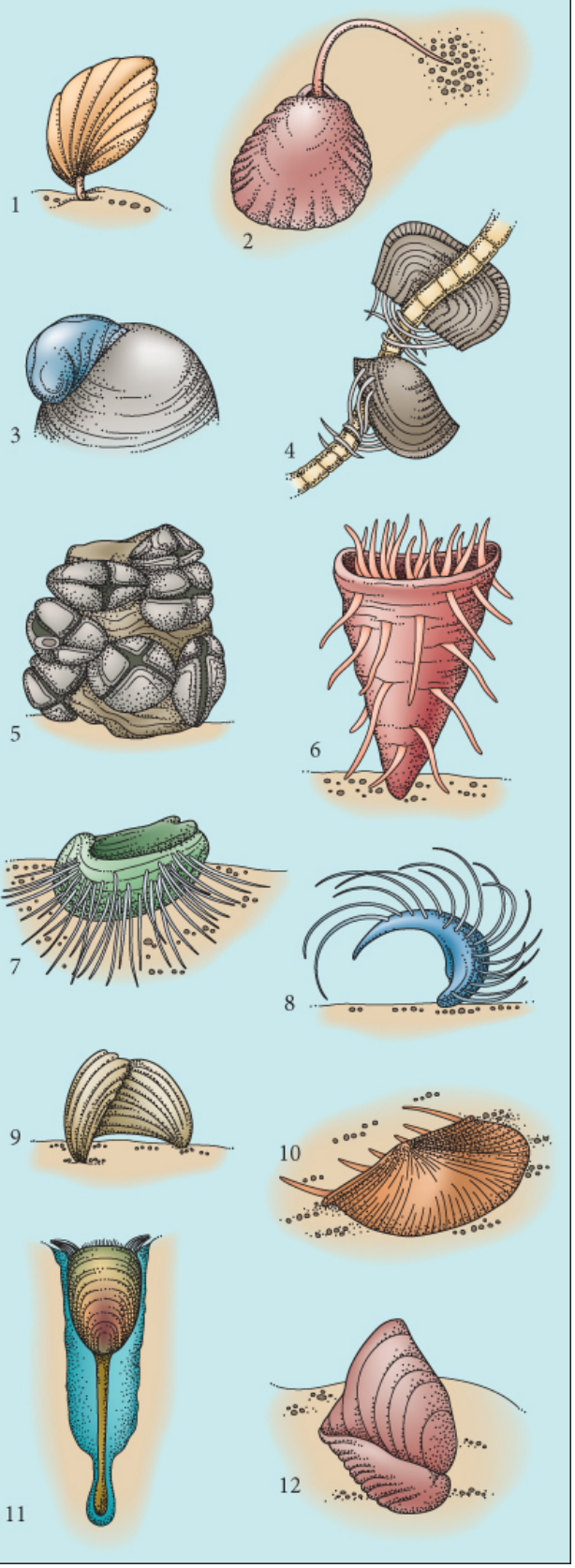
Brachiopod task 3: What is the sister group to the Brachiopoda, and of which clades is it a member?

Molluscs

Mollusc task 1: We also met molluscs in our lecture. Let's make sure we are able to identify the different groups in hand specimens. Look at the hand specimens in front of you, and for each, identify and note down which mollusc group each belongs to. For the cephalopods, using the image below left, try also to identify which subgroup the fossils may belong to. Make sure you can identify using the correct terminology the bits of the anatomy you can see!

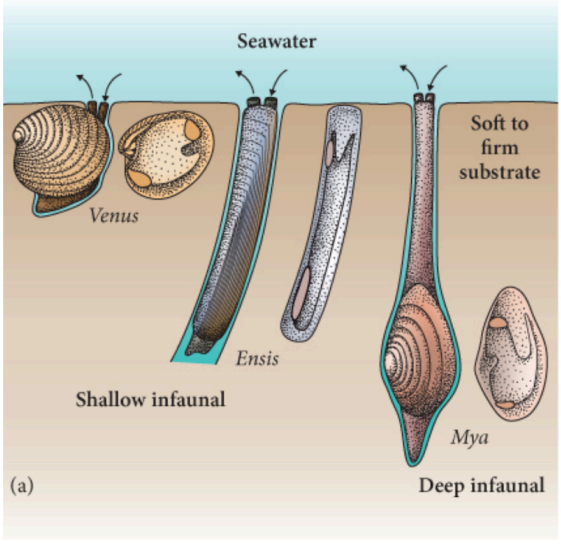


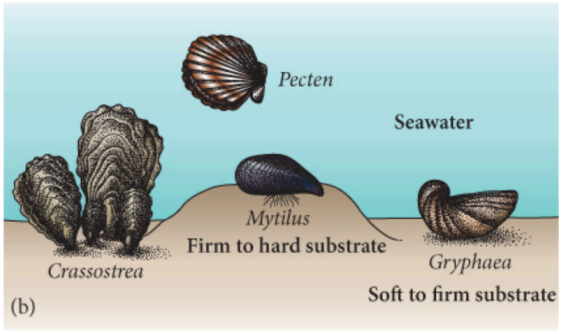
A1
Belemnites
Arctica
Pecten
Gryphea

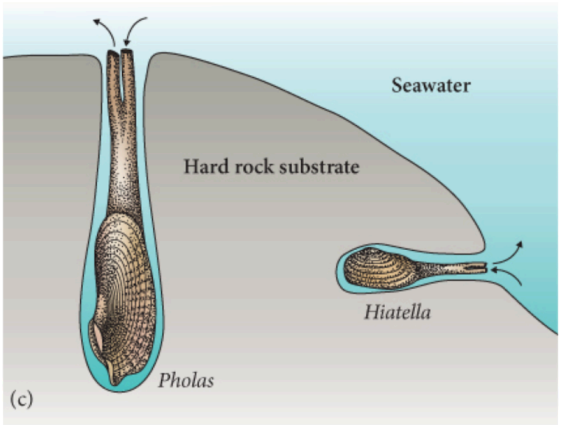
LIFESTYLE	BRACHIOPOD TAXA	ADAPTATIONS
Attached by pedicle Epifaunal – hard substrate (1) (plenipedunculate)	Orthides, rhynchonellides, spiriferides and terabratulides	
Epifaunal – soft substrate (2) (rhizopedunculate)	*Chlidonophora* and *Cryptopora*	
Cryptic	*Argyrotheca* and *Terebratulina*	
Interstitial	Acrotretides and *Gwynia*	
Cemented	*Craniops* and *Schuchertella*	
Encrusting (3)	Craniids and discinids	
Clasping spines (4)	*Linoproductus* and *Tenaspinus*	
Mantle fibers	Orthotetoids	
Unattached Cosupportive (5)	Pentamerids and trimerellids	
Coral-like (6)	Gemmellaroids and richthofeniids	
Recumbent	Strophomenides	
Pseudofaunal (7) and inverted (8)	*Waagenoconcha* and *Marginifera*	
Free-living (9, 10)	*Cyrtia, Chonetes, Neothyris* and *Terebratella*	
Mobile Infaunal (11)	Linguloids	
Semi-infaunal (12)	*Camerisma* and *Magadina*	

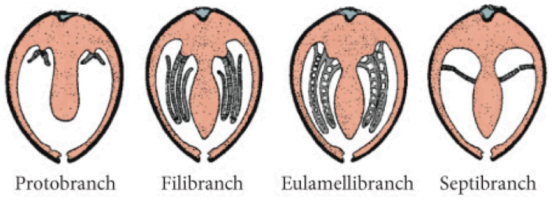
Molluscs task 2: Like we did with the brachiopods, we can think about the mode of life of these animals based on their shell shape, and use some common sense building from this. The sources below may help. Let's dive in!

Infaunal shallow burrowers e.g. *Glycimeris*		Equivalved, adductor muscles of equal sizes and commonly with strong external ornament
Infaunal deep burrowers e.g. *Mya*		Elongated valves, often lacking teeth and with permanent gape and a marked pallial sinus
Epifaunal with byssus e.g. *Mytilus*		Elongate valves with flat ventral surface and reduction of both the anterior part of the valve and the anterior muscle scar attached by thread-like byssus
Epifaunal with cementation e.g. *Ostrea*		Markedly differently shaped valves, sometimes with crenulated commissures; large single adductor muscle
Unattached recumbent e.g. *Gryphaea*		Markedly differently shaped valves, sometimes with spines for anchorage or to prevent submergence in soft sediment
Swimmers e.g. *Pecten*		Valves dissimilar in shape and size with very large, single adductor muscle and commonly with hinge line extended as ears
Borers and cavity dwellers e.g. *Teredo*		Elongate, cylindrical shells with strong, sharp external ornament; cavity dwellers commonly grow in dimly lit conditions following the contours of the cavity









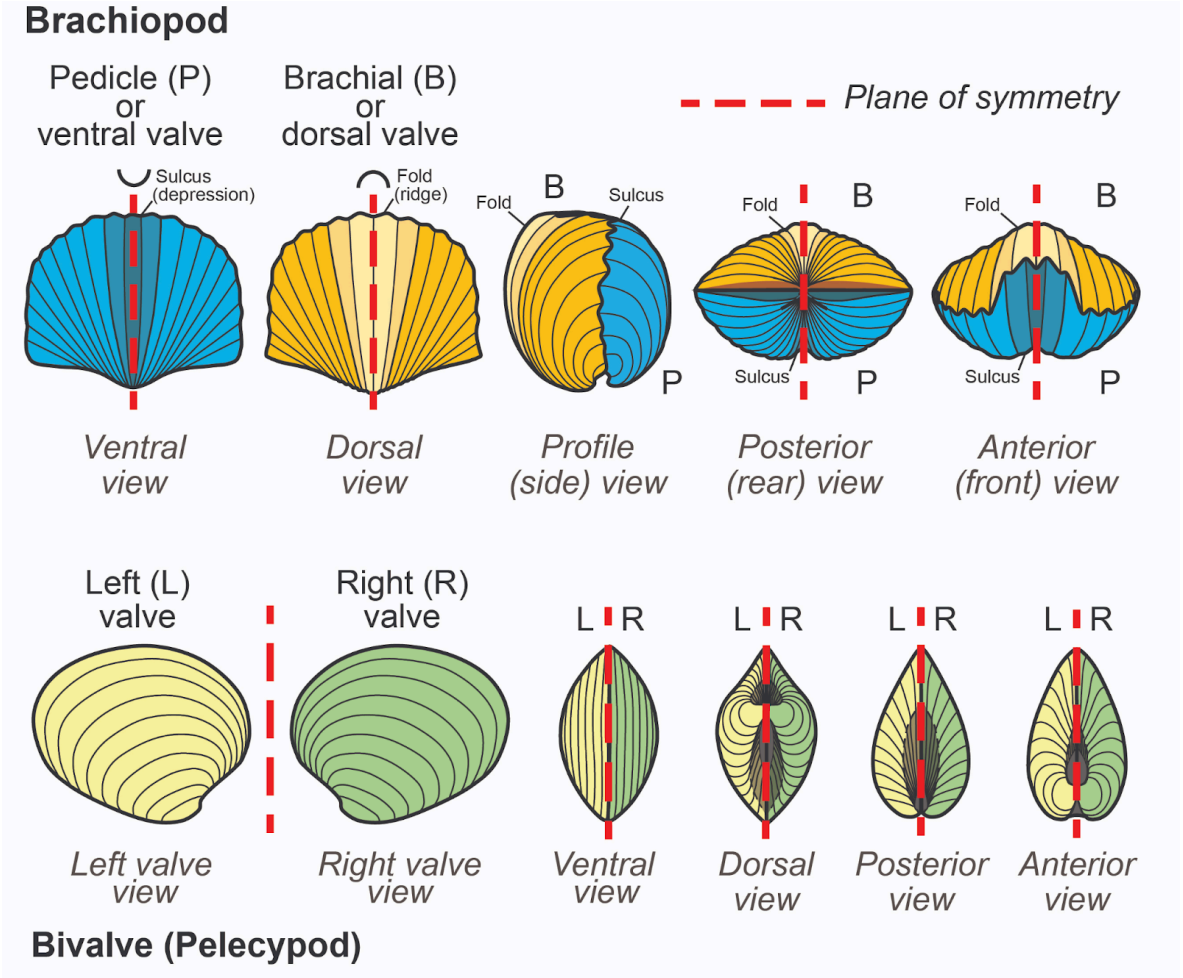
What was the mode of life of our fossils? How did they feed?

A1
Belemnites

Arctica
Pecten
Gryphea

Brachiopods vs bivalves

One of the harder bits of invertebrate palaeontology can be telling brachiopods and bivalves apart, especially when you have fragmented shells. So let's give this a go. Bear in mind that these are completely different animals from relatively distantly related animal groups (if you don't believe me have a look at the phylogeny I use in lectures). This is also reflected in major differences in their organisation. You may not have clocked this given the descriptions were separated by quite a few slides, so here's a little recap in the form of a figure (courtesy of the Kentucky Geological Survey):



So, for example, this makes it clear that in a brachiopod, one valve is dorsal, the other is ventral. In contrast in the bivalve, the shells represent the left, and the right hand side of the animal (if you're wondering how we know this, studying the *development* of the animal, and especially its embryo and early life stages, helps us identify across groups, the body axes. Developmental biology is *super cool*).

You can make sure you're happy with how to tell these groups apart using this part of the website, and then if you want to test yourself, there is a brachiopod v.s. bivalve quiz there I would encourage you to try:

Link *BvB*

B vs B task 1: Given this fact, *why* do you think it is that brachiopods and bivalves look so alike?

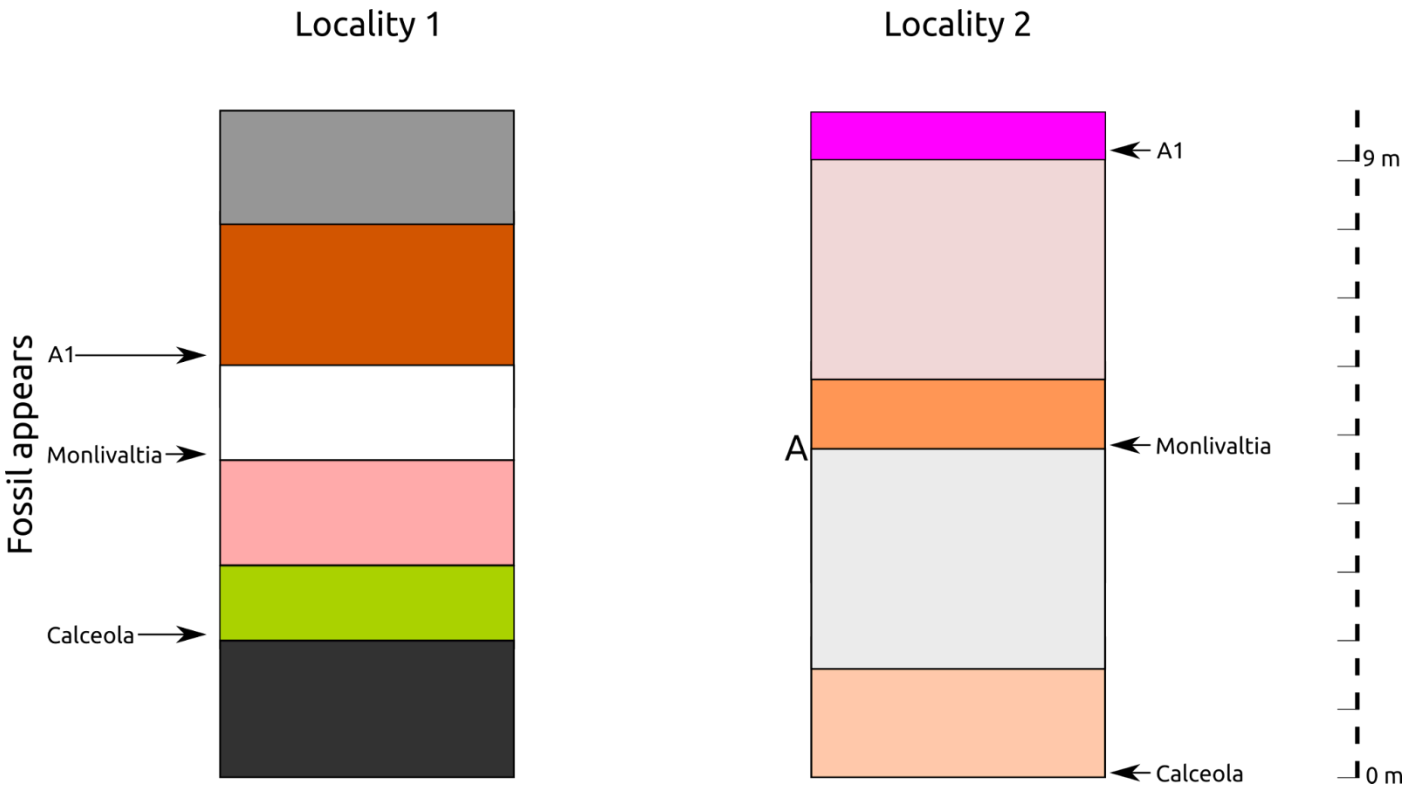
B vs B task 2: In front of you, you have two fossil bivalved organisms, A and B (check the image being projected at the front of the room to see which is which). Identify which is the brachiopod and which is the bivalve, and record why. Check with a GTA that you have got it correct.

A:
B:

Biostratigraphy 1

So we've learned a little about biostratigraphy. Let's try and put this into practice and draw on some isochrons. This may seem simple, but it's best to start with the simple tasks first. We'll ramp it up a little next week. Below you can see two stratigraphic sequences from two continents. Let's call them Oceania and Eurasia (we could even imagine Oceania had always been at war with Eurasia, if we so wished). The rocks are completely different lithologies (and hence different colours). But we can correlate lithological boundaries between the two in terms of time, using the appearance of some of the fossils we have already met.

Biostratigraphy task 1: The labels and arrows represent the appearance of particular fossils. Using these, mark on isochrons representing lithological boundaries of roughly equivalent age.



Biostratigraphy task 2: Cool, huh, but given that we know these lines represent equivalent ages, we can go a little further. See if you can work out, using the isochrons and the scale, the *relative* rate of sedimentation between Locality 1 and Locality 2. Do this for the period between the appearance of *Calceola* and A1.

Biostratigraphy task 3: You have hand specimens of *Calceola* (a Devonian rugose coral) and *Monlivaltia* (a Jurassic scleractinian coral) in front of you. Remind yourself of what these look like, and how they differ. For this exercise, let's assume that *Calceola* is present in every bed after its appearance until *Monlivaltia* appears. The label A on the left of the locality 2 column, marks an unconformity in these beds. What event in the history of life occurred between the deposition of the beds above and below this unconformity? You learned about it with Rob in Year 1. What minimum time must this unconformity represent? Justify your answer (you'll probably want to look at a geological column).